# Ch 9.2: Support Vector Classifier

## Lecture 27 - CMSE 381

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Wed, Nov 13, 2024

# Announcements

**Last time:**

- 9.1 Maximal Margin Classifier

**This lecture:**

- 9.2 Support Vector Classifier

**Announcements:**

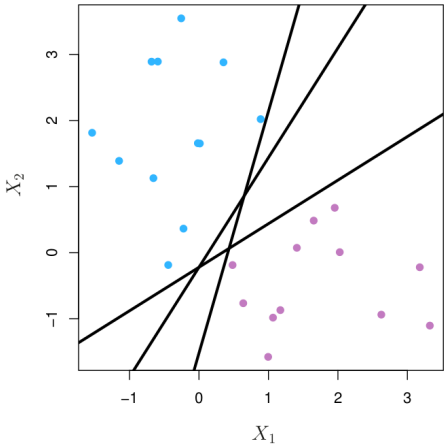
- HW #7 due Sunday 11/17

| Lec # | Date | | | Reading | HW |
|---|---|---|---|---|---|
| 21 | Mon | 10/28 | Polynomial & Step Functions | 7.1,7.2 | |
| 22 | Wed | 10/30 | Step Functions; Basis functions; Start Splines | 7.2 - 7.4 | |
| 23 | Fri | 11/1 | Regression Splines | 7.4 | HW #6 Due Sun 11/3 |
| 24 | Mon | 11/4 | Decision Trees | 8.1 | |
| 25 | Wed | 11/6 | Class Cancelled (Dr Munch out of town) | | |
| 26 | Fri | 11/8 | Random Forests | 8.2.1, 8.2.2 | HW #7 Due Sun 11/10 |
| 27 | Mon | 11/11 | Maximal Margin Classifier | 9.1 | |
| 28 | Wed | 11/13 | SVC | 9.2 | |
| 29 | Fri | 11/15 | SVM | 9.3, 9.4 | HW #8 Due Sun 11/17 |
| 30 | Mon | 11/18 | Single layer NN | 10.1 | |
| 31 | Wed | 11/20 | Multi Layer NN | 10.2 | |
| 32 | Fri | 11/22 | CNN | 10.3 | HW #11 Due Sun 11/24 |
| 33 | Mon | 11/25 | TBD: Unsupervised learning/clustering | 12.1, 12.4? | |
| | Wed | 11/27 | Virtual: Project office hours | | |
| | Fri | 11/29 | No class - Thanksgiving | | |
| | Mon | 12/2 | Review | | |
| | Wed | 12/4 | **Midterm #3** | | |
| | Fri | 12/6 | No class - EGR Design Day | | Project due |

Section 1

# Last time

# Separating Hyperplane




Require that for every data point:

$$\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip} > 0 \text{ if } y_i = 1$$
$$\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip} < 0 \text{ if } y_i = -1$$

*Equivalently*

Require that for every data point

$$y_i(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip}) > 0$$

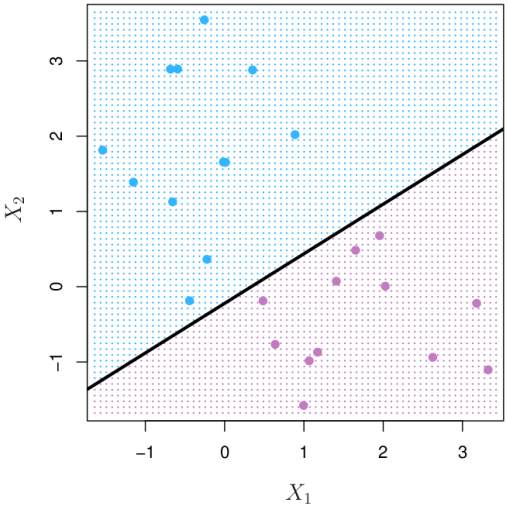
# Separating hyperplane becomes a classifier

If you have a separating hyperplane:

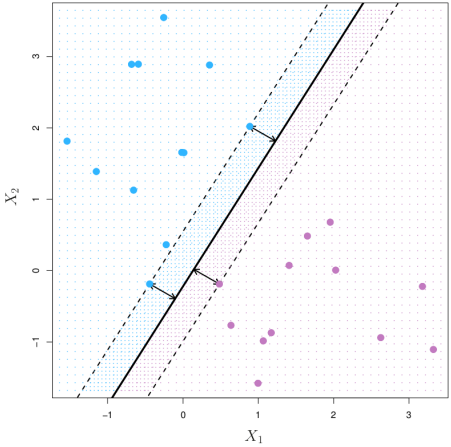
- Check

$$f(x^*) = \beta_0 + \beta_1 x_1^* + \beta_2 x_2^* + \cdots + \beta_p x_p^*$$

- If positive, assign $\hat{y} = 1$
- If negative, assign $\hat{y} = -1$



# Maximal margin classifier




- For a hyperplane, the *margin* is the smallest distance from any data point to the hyperplane.
- Observations that are closest are called *support vectors*.
- The *maximal margin hyperplane* is the hyperplane with the largest margin
- The classifier built from this hyperplane is the *maximal margin classifier*.

# Mathematical Formulation

$$\underset{\beta_0,\beta_1,\ldots,\beta_p,M}{\text{maximize}} \; M$$

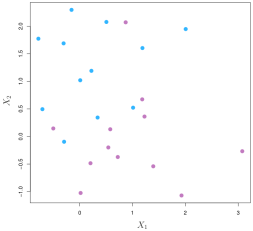
$$\text{subject to } \sum_{j=1}^{p} \beta_j^2 = 1,$$

$$y_i(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip}) \geq M \;\; \forall\, i = 1, \ldots, n$$

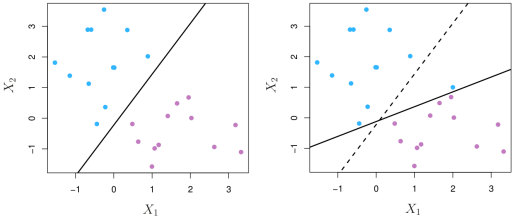
- Unit normal requirement, we can always write a given hyperplane this way.
- Last eq forces points to be on the correct side of the hyperplane in order for $M > 0$
- The product is the distance from the point to the hyperplane
- Making $M$ as big as possible is the maximal margin hyperplane

# Problems

## Might be no separating hyperplane




## Sensitivity to new points



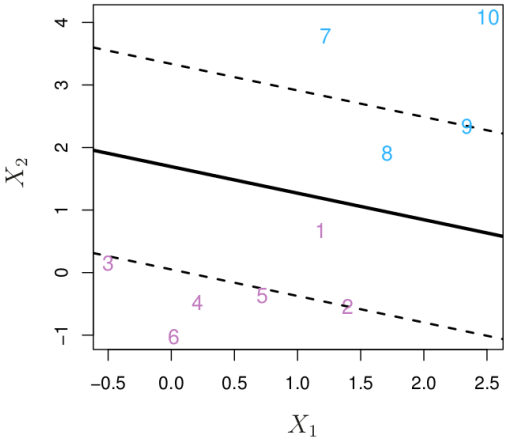
Section 2

# Support Vector Classifier

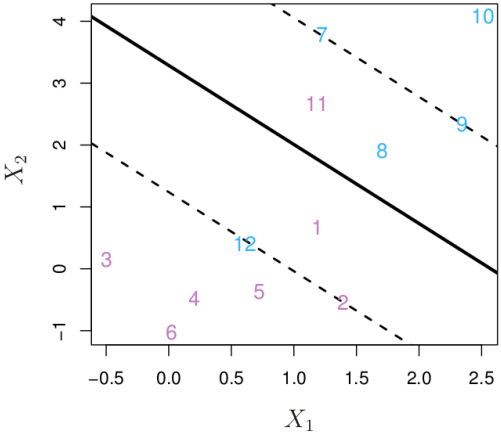
# Basic idea

- Be ok with having a classifier that isn't quite perfect
- Aim for greater robustness to individual observations
- Better classification of most of the training observations
- Result is a support vector classifier
- Soft margin classifier

# Soft margin




*Some points on wrong side of margin*




*Some points on wrong side of hyperplane (Misclassified)*

# Mathematical Formulation of SVC

$$\underset{\beta_0,\beta_1,\ldots,\beta_p,\epsilon_1,\ldots,\epsilon_n,\,M}{\text{maximize}} \quad M$$

$$\text{subject to} \quad \sum_{j=1}^{p} \beta_j^2 = 1,$$

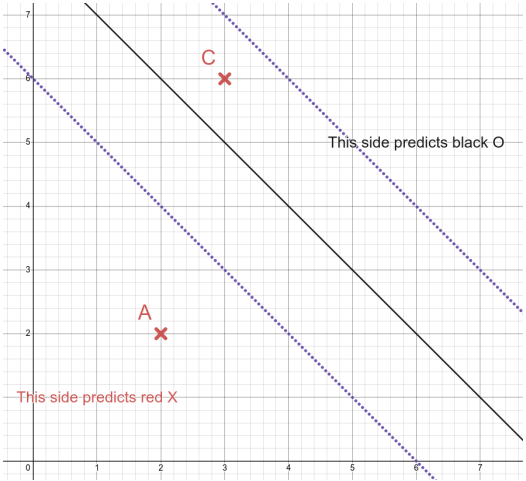
$$y_i(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip}) \geq M(1 - \epsilon_i),$$

$$\epsilon_i \geq 0, \quad \sum_{i=1}^{n} \epsilon_i \leq C,$$

- $C$ is nonnegative tuning parameter
- $M$ is the width of the margin
- $\varepsilon_1, \cdots, \varepsilon_n$ are slack variables allowing observations to go to the other side

# Find positive $\varepsilon$'s that will satisfy this

Fix $M = \sqrt{2}$ $\qquad y_i(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip}) \geq M(1 - \varepsilon_i)$

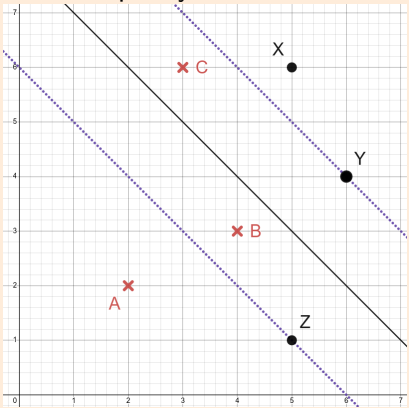



- Point out $M$ is distance from center hyperplane to noted margin.
- $A$ is on the correct side of the hyperplane.
  - Left side is already bigger than $M$, so set $\varepsilon_i = 0$.
- $C$ is on the wrong side of the hyperplane for its label.
  - Left side of the equation is negative. What $\varepsilon$ satisfies this?
  - $-\frac{\sqrt{2}}{2} = \sqrt{2}(1 - \varepsilon)$
  - $\varepsilon = 3/2$

# What is $\varepsilon$?

Fix $M = \sqrt{2}$ $\qquad y_i(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip}) \geq M(1 - \varepsilon_i)$

Fill in the table so that the inequality is satisfied.

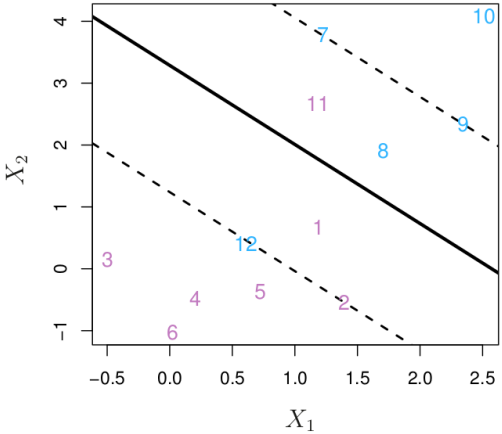



| *Point* | *Left Side* | $\varepsilon_i$ | $M(1-\varepsilon_i)$ |
|---|---|---|---|
| *A* | $2\sqrt{2}$ | *0* | $\sqrt{2}$ |
| *B* | $\frac{\sqrt{2}}{2}$ | $\frac{1}{2}$ | $\sqrt{2}(1-1/2) = \frac{\sqrt{2}}{2}$ |
| *C* | $-\frac{\sqrt{2}}{2}$ | 1.5 | $-\frac{\sqrt{2}}{2}$ |
| *X* | $\frac{3\sqrt{2}}{2}$ | *0* | $\sqrt{2}$ |
| *Y* | $\sqrt{2}$ | *0* | $\sqrt{2}$ |
| *Z* | $-\sqrt{2}$ | *2* | $\sqrt{2}(1-2) = -\sqrt{2}$ |

*To use for later slide: Total $\varepsilon$ is 4*

# What is $\varepsilon$?

- If $\varepsilon_i = 0$, then on correct side of margin
- If $\varepsilon_i > 0$ then on the wrong side of margin (Violated margin)
- If $\varepsilon_i > 1$ then on the wrong side of hyperplane



# What is $C$?

$$\underset{\beta_0,\beta_1,\ldots,\beta_p,\epsilon_1,\ldots,\epsilon_n,\,M}{\text{maximize}} \quad M$$

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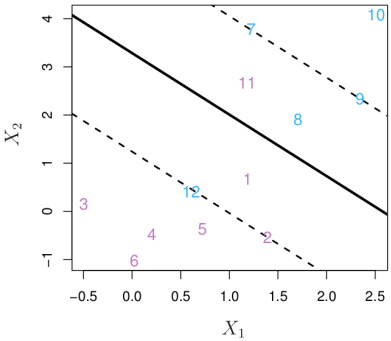
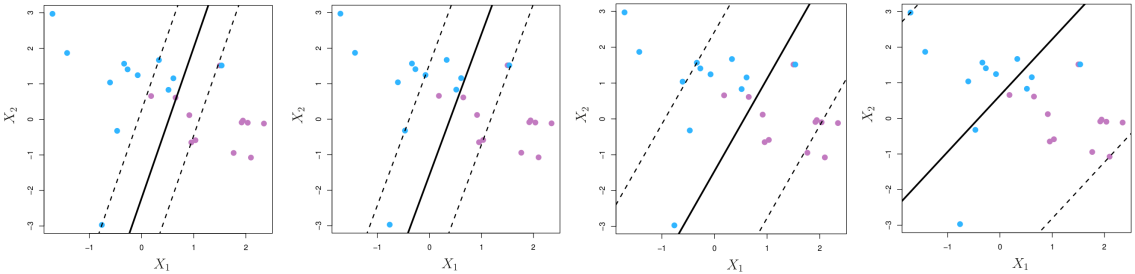
$$\epsilon_i \geq 0, \quad \sum_{i=1}^{n} \epsilon_i \leq C,$$

- Bounds sum of $\varepsilon_i$, so controls number & severity of violating margin (budget)
- $C = 0$ means no violations allowed
- $C > 0$ means at most $C$ observations can be on wrong side of hyperplane
- In previous example, our total of $\varepsilon$ was 4, so would be a valid hyperplane for $C$ at most 4.



# Examples messing with $C$




Increasing $C \rightarrow$

- For increasing $C$, we have more flexibility, so more points allowed violate margin/hyperplane
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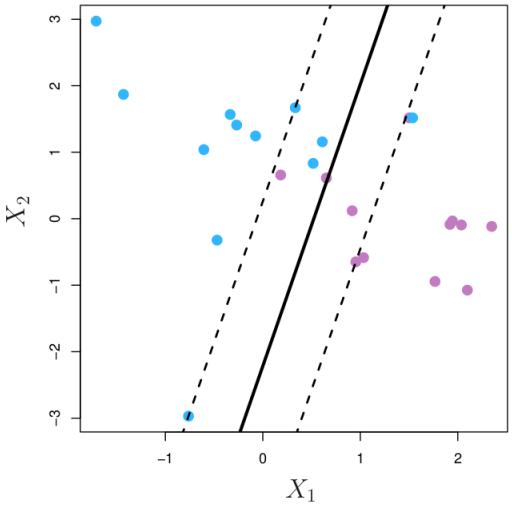
# What affects the hyperplane?




- Only observations on the margin or violating the margin affect the hyperplane
- These observations are called support vectors
- Changing other points positions doesn't affect hyperplane found
-

# Coding

# TL;DR

## Maximal Margin Classifier

$$\underset{\beta_0,\beta_1,\ldots,\beta_p,M}{\text{maximize}} \quad M$$

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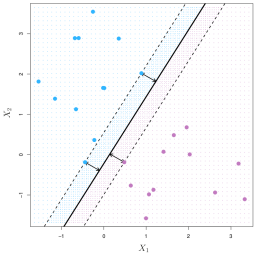


## Support Vector Classifier

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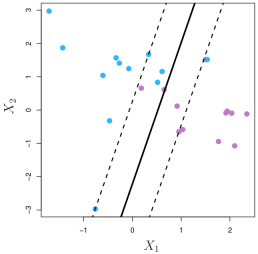

# Next time

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